

■ TWO POINTERS

Complete DSA Mentor Workbook

From Absolute Beginner → Competitive Programmer

20

Problems

5

Categories

Beginner → Story-Driven

Very Hard

Problems

*Master every Two Pointer pattern with hand-crafted story problems,
structured hints, and deep insights — no pattern left behind.*

AI DSA Mentor • 2024

■ How to Use This Workbook

This workbook contains 20 original problems across 5 progressive categories of the **Two Pointers** technique. Every problem is original, story-driven, and designed to teach a specific sub-pattern or insight. Solve them in order — each problem builds on the last.

Category	Pattern	Problems	Difficulty Range
1 — Opposite Ends	Classic 2-ptr on sorted array	TP-01 to TP-04	Beginner → Medium
2 — Same Direction	Slow/fast, sliding window	TP-05 to TP-08	Beginner → Medium
3 — Partition	Dutch flag, rearrangement	TP-09 to TP-11	Easy → Medium
4 — Strings	Palindrome, minimum window	TP-12 to TP-14	Beginner → Medium
5 — Advanced	4Sum, boats, longest palindrome	TP-15 to TP-20	Medium → Very Hard

When to Reach for Two Pointers

■ Sorted array + pair/triplet with target sum	Opposite ends
■ Find longest/shortest subarray with constraint	Sliding window / same-dir
■ In-place rearrangement without extra space	Partition
■ Palindrome check or expansion	String two pointers
■ Counting subarrays satisfying a condition	atMost(K) trick
■ Merging two sorted structures	Merge-style pointer

General Template: $i = 0$, $j = N-1$ (or $j = i+1$ for same direction). Move pointers based on a condition. Stop when $i \geq j$ (opposite) or $j == N$ (same-dir). Time complexity is almost always $O(N)$ or $O(N \log N)$ with a sort.

Category 1: Opposite-Ends Two Pointers

TP-01 The Royal Banquet Pairing

● Beginner

■ 1s | ■ 256MB

#two_pointers_opposite_ends

■ Story

King Aldric is hosting a grand banquet. He has N guests seated in a line, each with a known appetite score. The royal chef can only prepare one special dish that exactly feeds two guests simultaneously — the dish size equals the sum of their appetite scores. Given a target dish size T , find any two guests whose combined appetite equals T . The guests are already seated in increasing order of appetite (the king's seating rule).

■ Task

Given a sorted array A of N integers and a target T , find indices (1-indexed) of any two elements that sum to T . If no such pair exists, print -1.

■ Constraints

$2 \leq N \leq 2 \times 10^5$ | $1 \leq A[i] \leq 10^9$ | $1 \leq T \leq 2 \times 10^9$

■ Sample Test Cases

Input	Output	Note
5 9 1 3 5 7 8	2 4	$3+7=9$, guests 2 and 4
4 10 2 4 6 8	2 4	$4+6=10$
3 100 1 2 3	-1	No valid pair
2 2 1 1	1 2	Edge: minimum N

■ Hint (read only if stuck)

■ Place one pointer at the leftmost guest and one at the rightmost. If their sum is too large, move the right pointer left; if too small, move the left pointer right.

■ Key Insight

KEY INSIGHT: A sorted array + target sum → classic opposite-ends pattern. Time $O(N)$, Space $O(1)$. This is the foundation of ALL opposite-ends problems.

TP-02 The Wizard's Triplet Spell

● Easy

■ 2s | ■ 256MB

#two_pointers_three_sum

■ Story

Wizard Merinda needs to cast a stability spell using exactly three crystals whose combined power equals zero (positive and negative energies cancel). She has N crystals with integer power values. Find ALL unique triplets — no duplicate triplet should appear twice, even if crystals are identical. The order of crystals in a triplet does not matter.

■ Task

Given array A of N integers, find all unique triplets (a, b, c) with $a \leq b \leq c$ such that $a + b + c = 0$. Print the count on the first line, then each triplet.

■ Constraints

$3 \leq N \leq 3000$ | $-10^5 \leq A[i] \leq 10^5$

■ Sample Test Cases

Input	Output	Note
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6 -1 0 1 2 -1 -4	2 -1 -1 2 -1 0 1	Two distinct triplets
5 0 0 0 0 0	1 0 0 0	All zeros — one unique triplet
4 1 2 3 4	0	No negatives, no zero sum
3 -1000 0 1000	1 -1000 0 1000	Extreme values

- Hint (read only if stuck)
- Sort first. Fix the first element with a loop (skip duplicates!), then apply opposite-ends two pointers on the remaining subarray.
- Key Insight
- KEY INSIGHT: Reducing 3Sum to 2Sum via sorting + fixing one element is the core trick. Deduplication at each pointer is critical.

TP-03 The Harbour Container Stack

● Easy

■ 1s | ■ 256MB

#two_pointers_max_area

- Story
- Port master Elena oversees N vertical pillars along a harbour. She wants to choose two pillars to form the walls of a water reservoir. The amount of water the reservoir holds equals the distance between the pillars multiplied by the height of the shorter pillar. Find the maximum water the reservoir can hold. Elena cannot tilt or move pillars — only choose which two to use.

- Task
- Given N pillar heights, find the maximum water volume (width $\times$ min-height) achievable by choosing any two pillars.

- Constraints
- $2 \leq N \leq 10^5 \mid 1 \leq H[i] \leq 10^4$

■ Sample Test Cases

Input	Output	Note
9 1 8 6 2 5 4 8 3 7	49	Classic example: pillars 2 and 9 (heights 8,7), width 7
2 1 1	1	Minimum case
6 4 3 2 1 4	16	Both ends height 4, width 4
4 1 2 1 2	4	Multiple valid pairs

- Hint (read only if stuck)
- Start with widest possible (both ends). Gain from moving inward only if you move the shorter pillar's pointer.
- Key Insight
- KEY INSIGHT: Moving the taller pointer inward can never improve the answer (width decreases, height stays $\leq$ current short). Always advance the shorter-pillar pointer.

TP-04 The Alchemist's Closest Potion

● Medium

■ 2s | ■ 256MB

#two_pointers_closest_sum

- Story
- Alchemist Dorian is crafting a potion that needs exactly three ingredients with a combined potency as close to a target value P as possible. He has N ingredients sorted by potency. Find the triplet whose sum is closest to P . If two triplets are equally close, report the smaller sum.

Task

Given sorted array A of N integers and target P, find the triplet sum closest to P. Output that sum.

Constraints

$3 \leq N \leq 5000$ | $-10^4 \leq A[i] \leq 10^4$ | $-3 \times 10^4 \leq P \leq 3 \times 10^4$

Sample Test Cases

Input	Output	Note
4 2 -1 2 1 -4	2	-1+2+1=2, closest to 2
4 1 0 0 0 1	1	0+0+0=0 vs 0+0+1=1, both distance 1, 1 is exact
3 100 1 2 3	6	Only one triplet, sum=6
5 0 -4 -1 -1 0 1	-2	-4+2=-2 pair: -1-1+0=-2 closest to 0

Hint (read only if stuck)

Fix one element, use two pointers on the rest. Track closest sum seen so far.

Key Insight

KEY INSIGHT: Almost identical to 3Sum but instead of checking ==0, update a 'best answer' variable. Teaches you to adapt the pattern for 'closest' queries.

Category 2: Same-Direction Two Pointers (Slow & Fast / Sliding Window Foundation)

TP-05 The Plague Doctor's Quarantine

● Beginner

■ 1s | ■ 256MB

#two_pointers_remove_duplicates

Story

Plague doctor Silas must record only unique patient IDs in a scroll. The IDs arrive pre-sorted. He has limited ink and cannot use extra scrolls — he must overwrite the original list in-place, keeping only the first occurrence of each ID, and report how many unique patients there are. The remaining positions in the scroll are ignored.

Task

Given a sorted array A of N integers, remove duplicates in-place and return the count of unique elements. Modify A such that A[0..k-1] holds the k unique elements in order. Print k, then the first k elements.

Constraints

$1 \leq N \leq 10^5$ | $-10^9 \leq A[i] \leq 10^9$

Sample Test Cases

Input	Output	Note
6 1 1 2 3 3 4	4 1 2 3 4	4 unique elements
5 1 1 1 1 1	1 1	All same: 1 unique
4 -3 -1 0 5	4 -3 -1 0 5	Already unique, no change
1 42	1 42	Single element

Hint (read only if stuck)

Use a 'write' pointer (slow) and a 'read' pointer (fast). The slow pointer only advances when a new unique element is found.

Key Insight

KEY INSIGHT: The slow pointer marks the next write position; the fast pointer scans ahead. This slow/fast in-place overwrite pattern appears in dozens of problems.

TP-06 The Cartographer's Expedition

● Easy

■ 2s | ■ 256MB

#two_pointers_subarray_sum

■ Story

Cartographer Lyra is mapping a mountain range. She records the elevation change at each step as a positive integer. She wants to find the shortest contiguous stretch of terrain whose total elevation gain equals exactly a target G . Help her find that minimum length stretch, or report that no such stretch exists.

■ Task

Given array A of N positive integers and target G , find the minimum length contiguous subarray with sum exactly G . Print the length, or -1 if none.

■ Constraints

$1 \leq N \leq 10^5$ | $1 \leq A[i] \leq 10^4$ | $1 \leq G \leq 10^9$

■ Sample Test Cases

Input	Output	Note
8 7 2 3 1 2 4 3 1 2	2	$4+3=7$, length 2
5 15 1 2 3 4 5	5	$1+2+3+4+5=15$, entire array
3 100 1 2 3	-1	Max sum is 6, impossible
4 6 3 3 3 3	2	$3+3=6$, length 2

■ Hint (read only if stuck)

■ Expand the window by moving the right pointer. Shrink by moving the left pointer when sum equals target (looking for minimum).

■ Key Insight

KEY INSIGHT: With all-positive values, sum is monotonic as window grows/shrinks — this is what enables the two-pointer approach. Negative numbers would require prefix sums instead.

TP-07 The Spy Network's Code Breaker

● Easy

■ 2s | ■ 256MB

#two_pointers_k_distinct

■ Story

Spy coordinator Vance intercepts enemy messages encoded as arrays of integers. He wants to find the longest contiguous segment of the message that contains at most K distinct values (each value represents a different enemy agent ID). A longer segment means more context for decoding. Help him find the maximum such segment length.

■ Task

Given array A of N integers and integer K , find the length of the longest contiguous subarray containing at most K distinct values.

■ Constraints

$1 \leq N \leq 10^5$ | $1 \leq A[i] \leq 10^5$ | $1 \leq K \leq N$

■ Sample Test Cases

Input	Output	Note
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7 2 1 2 1 3 2 2 3	4	[1,2,1,3] or [3,2,2,3] both length 4
5 3 1 2 3 4 5	3	First 3 elements: 3 distinct
5 1 2 2 2 2 2	5	All same value: 1 distinct
6 2 1 2 3 1 2 3	4	[1,2,3,1] — wait, 3 distinct, try [1,2,3,1,2,3]→ no, answer is [2,3,1,2]→3+1=4

- Hint (read only if stuck)
- Key Insight
- Key Insight

TP-08 The Merchant's Profit Window

● Medium 2s | 256MB #two_pointers_exactly_k_distinct

- Story
- Task
- Constraints

■ Sample Test Cases

Input	Output	Note
5 2 1 2 1 2 3	7	7 subarrays with exactly 2 distinct
5 1 1 1 1 1 1	5	5 single-element subarrays (all same)
4 3 1 2 3 4	2	[1,2,3],[2,3,4] → 2
3 2 1 2 1	2	[1,2],[2,1],[1,2,1]→ 3? No: 1,2→2 and 2,1→2 and 1,2,1→2: answer 3. Verify.

- Hint (read only if stuck)
- Key Insight
- Key Insight

Category 3: Partition & Rearrangement Two Pointers

TP-09 The Arena's Color Sorting Trial

● Easy 1s | 256MB #two_pointers_dutch_flag

■ Story

Arena master Calix has N warriors, each wearing a red (0), white (1), or blue (2) sash. He must arrange them in order: all reds first, then whites, then blues — in a single pass without counting. The arrangement must be done in-place. This is the legendary Dutch Flag Trial.

■ Task

Given array A of N integers each 0, 1, or 2, sort it in-place in $O(N)$ time using at most $O(1)$ extra space. Print the sorted array.

■ Constraints

$1 \leq N \leq 3 \times 10^5 \mid A[i] \in \{0, 1, 2\}$

■ Sample Test Cases

Input	Output	Note
6 2 0 2 1 1 0	0 0 1 1 2 2	Standard case
5 0 0 0 0 0	0 0 0 0 0	All same (zeros)
5 2 2 2 2 2	2 2 2 2 2	All same (twos)
4 1 0 2 1	0 1 1 2	One of each + extra 1

■ Hint (read only if stuck)

■ Three pointers: low (next 0 pos), mid (current), high (next 2 pos). When $A[mid]=0$ swap with low++; when 2 swap with high--; when 1 just mid++.

■ Key Insight

KEY INSIGHT: Three-pointer partition (Dutch National Flag). The invariant: $A[0..low-1]=0$, $A[low..mid-1]=1$, $A[high+1..N-1]=2$ at all times.

TP-10 The Siege Engineer's Catapult Ranges

● Medium

■ 2s | ■ 256MB

#two_pointers_squares_sorted

■ Story

Siege engineer Brennan has N catapults sorted by their launch power (negative = fires backward, positive = fires forward). He needs to know, after squaring all launch powers (to compute kinetic energy), what the sorted order of energies would be. He must do this in $O(N)$ time — no time for re-sorting.

■ Task

Given a sorted (non-decreasing) array A of N integers (may include negatives), return the array of squares sorted in non-decreasing order. Do it in $O(N)$.

■ Constraints

$1 \leq N \leq 10^5 \mid -10^4 \leq A[i] \leq 10^4$

■ Sample Test Cases

Input	Output	Note
6 -4 -1 0 3 10	0 1 9 16 100	Negatives become large squares
4 -7 -3 2 3	4 9 9 49	$-7^2=49$ is largest
3 -5 -4 -3	9 16 25	All negative, reverse of input squares
3 1 2 3	1 4 9	All positive, squares in order

■ Hint (read only if stuck)

■ Largest squares come from the ends. Use two pointers at both ends, fill result array from right to left.

■ Key Insight

KEY INSIGHT: When merging from the extremes into the result array backwards, you avoid extra passes. This 'fill from end' trick is reusable in many merge scenarios.

TP-11 The Flood Model — Trapped Rainwater

● Medium

■ 2s | ■ 256MB

#two_pointers_trapping_rain

■ Story

Climate scientist Nora models a mountain terrain as an elevation map. After heavy rain, water gets trapped between peaks. She needs to compute the total volume of trapped rainwater. The terrain is given as a height array. Walls at both ends are implicitly zero. No water flows off the sides.

■ Task

Given array H of N non-negative integers representing the terrain height at each unit, compute total units of water trapped after rain.

■ Constraints

$1 \leq N \leq 3 \times 10^4 \mid 0 \leq H[i] \leq 10^4$

■ Sample Test Cases

Input	Output	Note
12 0 1 0 2 1 0 1 3 2 1 2 1	6	Classic example: 6 units trapped
6 4 2 0 3 2 5	9	Large valley: 9 units
4 1 0 1 0	1	Only 1 unit trapped
3 3 0 3	3	Symmetric valley

■ Hint (read only if stuck)

■ At each position, trapped water = $\min(\text{maxLeft}, \text{maxRight}) - H[i]$. Use two pointers to track maxLeft and maxRight without a separate pass.

■ Key Insight

KEY INSIGHT: The two-pointer approach avoids the $O(N)$ space of precomputed left/right max arrays. When $\text{leftMax} < \text{rightMax}$, process the left side (we know the limiting wall). This conditional processing based on which side has the smaller max is the deep insight.

Category 4: String & Palindrome Two Pointers

TP-12 The Oracle's Mirror Validation

● Beginner

■ 1s | ■ 256MB

#two_pointers_palindrome

■ Story

Oracle Thessaly receives prophecies written on stone tablets. A prophecy is considered 'mirrored' (palindrome) if it reads the same forwards and backwards, ignoring spaces and punctuation (only letters and digits count, and case is ignored). She has Q prophecies to validate before the eclipse.

■ Task

For each query string, determine if it is a valid palindrome (ignoring non-alphanumeric characters and case). Print YES or NO.

■ Constraints

$1 \leq Q \leq 100 \mid 1 \leq |S| \leq 10^5$

■ Sample Test Cases

Input	Output	Note
A man, a plan, a canal: Panama	YES	Classic palindrome
race a car	NO	Not palindrome
	YES	Empty after filtering: YES
No 'x' in Nixon	YES	Letters only: nixonnixon reversed

■ Hint (read only if stuck)

■ Filter only alphanumeric, lowercase. Then use left/right pointers moving inward.

■ Key Insight

KEY INSIGHT: Preprocessing the string into a clean form before applying two pointers is a common real-world pattern. Alternatively, skip invalid chars with the pointers themselves.

TP-13 The Forger's One Repair

● Easy

■ 1s | ■ 256MB

#two_pointers_palindrome_one_delete

■ Story

Forger Renata must make a document appear authentic. A document passes inspection if it is a palindrome. She can erase at most one character. Determine if the document can become a palindrome with at most one erasure.

■ Task

Given string S, determine if it can become a palindrome by removing at most one character. Print YES or NO.

■ Constraints

$1 \leq |S| \leq 10^5$ | S contains only lowercase English letters

■ Sample Test Cases

Input	Output	Note
abca	YES	Remove 'b' or 'c': 'aca' or 'aba'
raceacar	NO	No single removal works
abcba	YES	Already palindrome
deeee	YES	Remove one 'e': 'deee' is not... 'eeee' is! Remove 'd'

■ Hint (read only if stuck)

■ Use two pointers. When mismatch found, try skipping the left character OR the right character, and check if either remaining substring is a palindrome.

■ Key Insight

KEY INSIGHT: When you hit a mismatch with two pointers, you have exactly two choices to fix it. Trying both and verifying is O(N) total because the sub-check at most scans the rest once.

TP-14 The Linguist's Shortest Transform

● Medium

■ 2s | ■ 256MB

#two_pointers_minimum_window

■ Story

Linguist Evander studies ancient manuscripts. He has a source text T and a pattern word P. He needs to find the shortest contiguous window in T that contains all characters of P (in any order, with multiplicity). His translation machine can only process a window, not the full text.

■ Task

Given strings T (text) and P (pattern), find the minimum length window in T that contains all characters of P. Output the window substring. If none, print -1.

■ Constraints

$1 \leq |P| \leq |T| \leq 10^5$ | Both strings contain only lowercase letters

■ Sample Test Cases

Input	Output	Note
adobecodebanc abc	banc	Classic example
aaabbbccc abc	abbbc	Multiple valid windows, 'abbbc' is shortest
xyz abc	-1	No 'a','b','c' all present
a a	a	Single char match

■ Hint (read only if stuck)

■ Expand right until all chars of P are covered; then shrink from left as much as possible. Track coverage with a count variable.

■ Key Insight

KEY INSIGHT: The 'formed' counter (how many unique characters in P are satisfied with required frequency) lets you check window validity in $O(1)$. The two-pointer shrink-expand runs in $O(|T|)$.

Category 5: Advanced & Mixed Two Pointer Patterns

TP-15 The Botanist's Fruit Basket

● Medium

■ 2s | ■ 256MB

#two_pointers_at_most_2_distinct

■ Story

Botanist Iris travels a fruit garden where each tree bears one type of fruit (represented by an integer). She carries two baskets, each holding only one fruit type. Starting from any tree, she picks fruit from consecutive trees until she must stop (she cannot put a third fruit type in her baskets). Find the maximum fruits she can collect in one journey.

■ Task

Given array A of N integers (fruit types), find the maximum length of a contiguous subarray with at most 2 distinct values.

■ Constraints

$1 \leq N \leq 10^5$ | $1 \leq A[i] \leq 10^5$

■ Sample Test Cases

Input	Output	Note
5 1 2 1 2 3	4	$[1,2,1,2] = 4$
6 0 1 2 2 2 1	5	$[1,2,2,2,2]$ wrong... $[0,1,2,2,2] \rightarrow 3$ distinct. $[1,2,2,2]=4$. $[1,2,2,2,1]=5!$
4 3 3 3 3	4	All same, entire array
7 1 2 3 2 2 1 4	4	$[2,3,2,2] \rightarrow 3$ distinct, $[3,2,2,1]$ ok len 4

■ Hint (read only if stuck)

■ Same pattern as TP-07 with $K=2$. A HashMap tracks at most 2 types; shrink when a third appears.

■ Key Insight

KEY INSIGHT: This is just 'at most K distinct' with $K=2$ — demonstrating how one pattern solves many themed problems. Recognizing the underlying pattern is the skill.

TP-16 The General's Subarray Battle Count

● Medium

■ 2s | ■ 512MB

#two_pointers_count_subarrays_condition

■ Story

General Maxis needs to deploy squads. Each squad is a contiguous subarray of soldier power values. A squad is 'combat-ready' if the product of its soldiers' powers is strictly less than a threshold K . Count all possible combat-ready squads. All power values are positive.

■ Task

Given array A of N positive integers and integer K , count the number of contiguous subarrays whose product of elements is strictly less than K .

■ Constraints

$1 \leq N \leq 3 \times 10^4$ | $1 \leq A[i] \leq 1000$ | $0 \leq K \leq 10^6$

■ Sample Test Cases

Input	Output	Note
4 100 10 5 2 6	8	$[10],[5],[2],[6],[10,5],[5,2],[2,6],[5,2,6] \rightarrow 8$ subarrays
3 0 1 2 3	0	$K=0$: no product < 0 with positives
4 1 1 1 1 1	0	$K=1$: product ≥ 1 always
5 10 1 2 3 4 5	7	Count subarrays with product < 10

■ Hint (read only if stuck)

■ Maintain a sliding window product. For each right pointer position, the number of valid subarrays ending at right is $(right - left + 1)$.

■ Key Insight

KEY INSIGHT: When you fix the right boundary and find the smallest left such that the window is valid, ALL subarrays $[left..right]$, $[left+1..right]$, ..., $[right..right]$ are valid. Adding $(right-left+1)$ counts them all in $O(1)$.

TP-17 The Architect's Longest Valid Bridge

● Hard

■ 3s | ■ 256MB

#two_pointers_longest_ones_with_k_flips

■ Story

Architect Celeste is designing a bridge. The structural plan is a binary string: 1 = strong pillar, 0 = weak support. She has budget to reinforce at most K weak supports (flipping 0s to 1s). Find the longest contiguous stretch of the bridge that can be made entirely strong (all 1s) using at most K flips.

■ Task

Given binary array A of N integers (0s and 1s) and integer K , find the maximum length of a contiguous subarray that contains at most K zeros.

■ Constraints

$1 \leq N \leq 10^5$ | $A[i] \in \{0, 1\}$ | $0 \leq K \leq N$

■ Sample Test Cases

Input	Output	Note
6 2 1 1 1 0 0 0	5	Flip last 2 zeros: [1,1,1,0,0,0]→[1,1,1,1,1,0] len 5
10 2 1 1 0 0 0 1 1 0 1 1	6	Best window: [1,1,0,0,0,1,1]→flip 3 is too many; [0,1,1,0,1,1]→flip 2: len 6
5 0 0 0 1 1 0	2	K=0: max run of 1s = 2
5 5 0 0 0 0 0	5	K=5: entire array

■ Hint (read only if stuck)

■ Sliding window. Track count of zeros in window. When zeros exceed K, move left pointer past the first zero in the current window.

■ Key Insight

KEY INSIGHT: Instead of tracking the leftmost zero explicitly, you can use a 'lazy shrink' — only shrink the window when its size would NOT improve the answer. This gives an elegant O(N) solution where the window never contracts, only shifts.

TP-18 The Mathematician's Missing Ranges

● Hard

■ 3s | ■ 256MB

#two_pointers_four_sum

■ Story

Mathematician Quintus studies number quartets. He has a sequence of N integers and a target T. He needs all unique quadruplets (a, b, c, d) such that $a + b + c + d = T$. As a researcher, he must list them without duplicates, sorted lexicographically.

■ Task

Given array A of N integers and target T, find all unique quadruplets ($a \leq b \leq c \leq d$) with sum T. Print count on first line, then each quadruplet.

■ Constraints

$4 \leq N \leq 200 \mid -10^9 \leq A[i] \leq 10^9 \mid -10^9 \leq T \leq 10^9$

■ Sample Test Cases

Input	Output	Note
6 0 1 0 -1 0 -2 2	3 -2 -1 1 2 -2 0 0 2 -1 0 0 1	3 unique quadruplets
5 0 0 0 0 0 0	1 0 0 0 0	All zeros
4 4 1 1 1 1	1 1 1 1 1	Only one quadruplet
4 100 1 2 3 4	0	Sum is 10, not 100

■ Hint (read only if stuck)

■ Fix two outer elements with loops (skip duplicates at each), then apply 3Sum's two-pointer technique on the remaining subarray.

■ Key Insight

KEY INSIGHT: kSum generalizes: each outer loop reduces k by 1 until you reach 2Sum (two pointers). For 4Sum: O(N³). Deduplication logic is the tricky part — always skip when $i > \text{start}$ and $A[i] == A[i-1]$.

TP-19 The Shipwright's Greedy Pairs

● Hard

■ 3s | ■ 256MB

#two_pointers_boats_to_save_people

■ Story

Shipwright Cassius must rescue N people from a sinking island. Each rescue boat holds at most 2 people and has a weight limit W . Each person has a weight. Find the minimum number of boats needed to save everyone. Cassius wants to pair heavy and light people optimally.

■ Task

Given array A of N people's weights and limit W (each $A[i] \leq W$ guaranteed), find the minimum number of boats needed where each boat holds ≤ 2 people with total weight $\leq W$.

■ Constraints

$1 \leq N \leq 5 \times 10^4 \mid 1 \leq A[i] \leq W \leq 3 \times 10^4$

■ Sample Test Cases

Input	Output	Note
4 3 1 2 2 3	3	$(1,2), (2), (3) = 3$ boats
3 3 3 2 2	3	$(3), (2), (2) = 3$ boats, no pairing helps
5 5 3 2 2 1 1	3	$(3,2), (2,1), (1) \rightarrow 3$; or $(3,1), (2,1), (2) \rightarrow 3$
2 10 5 5	1	$5+5=10 \leq 10$: one boat

■ Hint (read only if stuck)

■ Sort. Greedily pair the heaviest with the lightest (if they fit). If not, the heaviest goes alone.

■ Key Insight

KEY INSIGHT: Sort + greedy two-pointer works because the optimal solution never benefits from pairing two heavy people — the lightest person is always the best pairing candidate for the heaviest.

TP-20 The Clockmaker's Circular Palindrome

● Very Hard

■ 4s | ■ 256MB

#two_pointers_longest_palindromic_substring

■ Story

Clockmaker Theo engraves palindromic motifs on clock faces. He has a string of N characters and wants to find the longest palindromic substring — the longest contiguous part of the string that reads the same forwards and backwards. This motif will be the centrepiece of his masterwork clock.

■ Task

Given string S of length N , find the longest palindromic substring. If multiple answers of the same maximum length exist, output the one that appears first. Output the substring and its length.

■ Constraints

$1 \leq N \leq 10^3 \mid S$ contains only lowercase English letters

■ Sample Test Cases

Input	Output	Note
babad	bab 3	'bab' or 'aba' both valid, output first
cbbd	bb 2	'bb' is longest palindrome
a	a 1	Single char is palindrome
racecarxyz	racecar 7	racecar is the full palindromic prefix

■ Hint (read only if stuck)

■ Expand around every center. Each character is an odd-length center; each gap between characters is an even-length center. Total: $2N-1$ centers.

■ **Key Insight**

KEY INSIGHT: The 'expand around center' technique uses two pointers expanding outward (not inward). For each of $2N-1$ centers, expand while chars match. $O(N^2)$ time, $O(1)$ space — elegant and interview-standard.

■ What's Next?

Congratulations on completing the Two Pointers Workbook! You've mastered: opposite-end pointers, slow/fast pointers, partition techniques, string palindrome patterns, sliding window with constraints, and advanced combinations like 4Sum and trapping rain water.

Next Pattern	Key Connection to Two Pointers
Binary Search	Sorted array + target → but finding count/position, not pair
Sliding Window (advanced)	Two pointers IS sliding window for contiguous subarrays
Prefix Sums	Enables two pointers on problems with negative numbers
Merge Sort / Merge Step	Merge uses two pointers on two sorted arrays
Monotonic Stack	When two pointers need $O(1)$ range min/max lookups

Keep solving. Keep recognising patterns. The best competitive programmers don't memorise solutions — they see the underlying structure instantly. You now have the foundation.